



STUDY

European Logistics Tech: Durable Margin Analysis

A deep dive into the European logistics tech sector, focusing on the identification of sustainable margins for investors, and providing data-backed conclusions essential for informed investment decisions.

STUDY REPORT

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• Executive Summary

Finding 1 — European logistics tech is in a strong expansion phase

The market is being reshaped by large transportation and warehousing spend intersecting with fragmented software and automation solutions. Growth is being driven by rising investment in compliance, resilience, automation, and data orchestration, supported by a B2C e-commerce market valued at €930 billion in 2024.

Finding 2 — WMS and warehouse automation stand out as scalable investment areas

Investment opportunities span TMS, WMS, freight visibility, last-mile, digital brokerage, and warehouse automation, but WMS and warehouse automation are specifically highlighted as highly scalable in Europe. Demand is reinforced by sustained e-commerce growth and increasing interest in warehouse automation and robotics.

Finding 3 — Regulation is shifting from headwind to product opportunity

Regulatory catalysts such as the Data Act, CountEmissionsEU, and EU ETS are accelerating adoption and shaping both product design and pricing. Vendors that embed compliance into workflows are better positioned to monetize demand, particularly in regulation-heavy processes and multimodal carbon-aware planning.

Finding 4 — Competitive advantage is moving to scaled workflow orchestrators

At the application layer, the market is shifting toward scaled platforms that combine network reach, proprietary data, and embedded workflows. Solutions with execution rights and compliance workflows have a stronger position than standalone visibility tools, while consolidation is increasing capability bundling pressure on smaller vendors.

Finding 5 — Margin quality depends more on business model than end-market growth

The strongest margins are associated with execution-critical SaaS and network platforms, while transaction-led and hardware-heavy models face constraints from pass-through costs. Asset-light SaaS models are therefore more attractive, especially where deep ERP and logistics integration creates high switching costs and durable retention.

Finding 6 — TMS and WMS are identified as investment-grade, despite clear execution risks

The summary explicitly identifies TMS and WMS as investment-grade segments, with investment criteria centered on margin durability, operational embedment, and compliance-driven monetization. Key risks remain commoditization, integration complexity, and regulatory burden, although these can be mitigated when compliance requirements are turned into productized modules.

1

Sector Landscape & Investment Context

1

Sector Landscape & Investment Context

European logistics tech sits at the intersection of a very large underlying transport and warehousing spend base and a still-fragmented software and automation layer. For investors, the key point is that **the addressable market is expanding faster than core freight volumes** because compliance, resilience, automation, and data orchestration are increasing technology spend per shipment, per warehouse, and per supplier relationship.^{22 29 52}

The investable universe spans TMS, WMS, freight visibility and control towers, last mile software, digital brokerage, and warehouse automation. Across these categories, **WMS and warehouse automation remain the clearest scale markets in Europe today, while visibility, emissions accounting, and orchestration layers are the fastest-improving strategic control points** as EU regulation and supply chain redesign force customers to digitise execution and reporting.^{40 21 46}

1.1

Market Sizing and Growth Projections

Title — European logistics tech market snapshots 2023 to 2026

SEGMENT	2023 TO 2026 MARKET VIEW	GROWTH SIGNAL
European B2C e commerce turnover	€887 billion in 2023, €930 billion in 2024. Western Europe represented 57% of 2024 turnover at €467 billion.	Positive demand base for parcel, last mile, WMS, and fulfilment tech.
EU business e sales penetration	16% of total business turnover in 2015, rising to 19% in 2025.	Structural digitisation of order flows and fulfilment complexity.
Europe warehouse automation	\$5.76 billion in 2025, \$6.79 billion in 2026.	17.86% CAGR for 2026 to 2031.
Europe warehouse robotics	2026 market expected to enter a high growth phase.	15.5% CAGR for 2026 to 2033.
Europe WMS	Third party estimates place 2025 market at \$1.56 billion.	Market research sources indicate double digit growth into 2030 plus.

Source: ^{26 22 46 44 32}

- Market framing: The most defensible way to size the sector is to anchor on the underlying European logistics and e commerce flow base rather than rely on a single headline “logistics tech” number. European B2C e commerce reached €930 billion in 2024, up from €887 billion in 2023, while EU businesses reported e sales equal to 19% of turnover in 2025, both of which expand the software and automation layer attached to fulfilment and transport operations.^{26 22}
- WMS: Europe WMS is already a meaningful software category, with third party market estimates placing the market at about \$1.56 billion in 2025 and Germany alone at roughly \$898.7 million in 2025 in one 2026 market publication. The exact absolute figure varies by methodology, but the directional conclusion is consistent across sources: **WMS is a scaled, multi billion euro**

category with sustained double digit growth driven by omnichannel complexity and warehouse automation integration ^{32 28}

- Warehouse automation and robotics: This is the clearest high growth subsector in the European stack. Mordor estimates Europe warehouse automation at \$5.76 billion in 2025 and \$6.79 billion in 2026, implying strong near term expansion, while Grand View projects 15.5% CAGR for European warehouse robotics from 2026 to 2033. ^{46 44}
- TMS and transport execution software: Gartner's March 30, 2026 Magic Quadrant confirms that the TMS market is growing and becoming more complex, but public Europe only market sizing is less transparent than for WMS and automation. For investors, that opacity matters less than the commercial reality that TMS budgets are being pulled by multimodal complexity, carrier procurement pressure, and the need to integrate emissions and resilience reporting into transport workflows. ^{40 21}
- Freight visibility and control towers: Reliable Europe specific market size disclosures are limited, so a precise 2023 to 2026 revenue series could not be verified from primary public sources. What is verifiable is that resilience and due diligence requirements are increasing demand for network monitoring and orchestration layers, especially where platforms can combine shipment visibility, supplier mapping, and emissions accounting in one workflow. ^{31 11}
- Last mile and digital brokerage: Public Europe specific market sizing is also fragmented here, but parcel and road freight software demand is supported by continued parcel volume growth and a difficult carrier cost environment. DHL reported that parcel growth remained a structural feature in 2025 and that European road freight demand was subdued while costs stayed high, which typically increases buyer appetite for routing, procurement, and network optimisation tools. ^{42 41}
- Investable universe mapping: The most relevant European categories for deal teams are TMS, WMS, freight visibility and control towers, last mile delivery software, digital freight brokerage, and warehouse automation and robotics. **From a market growth perspective, automation and WMS offer the clearest top down expansion, while visibility, control tower, and emissions layers offer the strongest strategic adjacency to regulation and resilience budgets** ^{46 21 40}

1.2

Structural Tailwinds

- Nearshoring and regionalisation: The European Commission has explicitly identified ongoing reallocation of EU imports toward reshoring, nearshoring, and partnershoring as supply chains are reconfigured for resilience and economic security. That shift increases network complexity rather than reducing it, which supports spend on TMS, supplier visibility, customs workflow, and control tower software. ^{29 31}
- E commerce as a persistent demand engine: European B2C e commerce reached €930 billion in 2024, and EU businesses said e sales represented 19% of turnover in 2025. This matters because e commerce drives the most software intensive logistics use cases, including slotting, labour planning, returns, parcel orchestration, and dynamic last mile routing. ^{26 22}
- ESG and emissions compliance: CountEmissionsEU entered into force on June 1, 2026 and establishes an EU wide methodology aligned with EN ISO 14083:2023 for transport emissions accounting. Even though disclosure is initially voluntary, the regulation standardises the data model for freight emissions, which creates a direct product opportunity for logistics software

vendors that can embed auditable carbon calculation, reporting, and optimisation into transport workflows.²¹

- Post-COVID resilience budgets: The resilience agenda has moved from contingency planning to operating model redesign. Commission survey evidence from 2025 shows firms responding to foreign market tensions through digitisation, enhanced monitoring, and in some cases relocation of production hubs, all of which increase demand for orchestration software and scenario planning tools.³¹
- Due diligence and traceability: The policy direction in Europe is toward deeper supply chain accountability, not lighter reporting. Research modelling 30 million EU firms and 900 million supply links found that 82.4% of EU companies are likely to have child or forced labour exposure by the second tier and more than 99.1% by the third tier, underscoring why supplier mapping, traceability, and compliance by design are becoming budgeted capabilities rather than optional analytics.¹¹
- Labour cost and automation pressure: DHL stated that in the European road freight market, costs remained high in 2025 due to rising labour costs, difficult market conditions, and tighter capacity. That is a direct tailwind for warehouse automation, route optimisation, and procurement software because customers increasingly need technology to offset wage inflation and improve asset utilisation.^{41 46}
- Technology readiness: Recent arXiv work in warehouse orchestration and robotic picking shows that AI and reinforcement learning are moving from concept to operational tooling in logistics environments, especially in warehouse execution. These papers are not market sizing sources, but they support the view that the technology stack is mature enough for commercial deployment, which strengthens the case for investing in software plus automation platforms rather than point solutions.^{1 2 3}

1.3

Investment Timing Rationale

- Regulatory catalyst is immediate, not theoretical: The Data Act has applied since September 12, 2025, and CountEmissionsEU entered into force on June 1, 2026. Together they increase the value of platforms that can capture operational data, standardise it, share it securely across counterparties, and convert it into compliance grade outputs.^{24 21}
- Macro backdrop favours productivity software: DHL expects the contract logistics market to continue its long term growth trend in 2026 and expects national and pan European e commerce to continue steadily, even as freight markets remain operationally challenging. That combination of modest volume growth and persistent cost pressure is typically the best environment for logistics tech adoption because buyers need measurable productivity gains rather than speculative transformation projects.^{52 41}
- Capex timing in automation is favourable: Europe warehouse automation is estimated to rise from \$5.76 billion in 2025 to \$6.79 billion in 2026, indicating that customers are already committing budget despite a mixed freight cycle. For investors, this is important because it suggests automation spend is being justified by structural labour and service level economics rather than by a short lived freight boom.⁴⁶
- Data standardisation is creating new software categories: CountEmissionsEU aligns transport emissions accounting to a common methodology, while the Data Act pushes toward fairer

access to industrial and connected product data. **This is a classic category creation moment** for vendors that can turn fragmented logistics data into auditable workflows for procurement, reporting, and optimisation.^{21 24}

- Resilience spending is now board level: The Commission's 2025 evidence on business adjustment to foreign market tensions shows that firms are actively changing sourcing and production strategies, including relocation and digitisation. That means logistics tech is increasingly funded from strategic resilience and compliance budgets, not only from line item IT budgets, which improves sales urgency for mission critical platforms.³¹
- Why now for investors: The sector is entering a period where regulation, cost inflation, and network redesign are all forcing adoption at once. **That combination is especially attractive for deal teams because it tends to separate durable workflow platforms from discretionary point tools, allowing capital to target the parts of the stack where pricing power and retention can improve as compliance and execution become more tightly coupled**^{40 21 52}
- Source limitation note: Precise 2023 to 2026 Europe only market size series for TMS, freight visibility, last mile software, and digital brokerage could not be verified from primary public disclosures within the available search set. The section therefore uses verified public data for the largest adjacent categories and clearly labels third party estimates where used, rather than overstating precision.

2

Competitive Structure & Moat Mapping

2

Competitive Structure & Moat Mapping

European logistics tech remains structurally fragmented at the application layer, but competitive power is concentrating around a small set of scaled orchestrators that combine physical network reach, proprietary operating data, and embedded workflows. **The investable conclusion is that durable advantage is strongest where software is tied to execution rights, carrier connectivity, customs or compliance workflows, or warehouse process control, and weakest where vendors sell stand-alone visibility or rate comparison without owning a deeper system of action.** 78 71 94 96

Recent consolidation has reinforced this pattern. DSV’s Schenker integration, Kuehne+Nagel’s network expansion through TDN, DHL’s targeted healthcare and returns acquisitions, Descartes’ continued roll-up of TMS and final-mile software, and project44’s move from visibility into agentic orchestration all point to the same market direction: fewer broad platforms, more capability bundling, and rising pressure on subscale point solutions. **White space remains in Europe where regulation-heavy workflows, SME digitisation, and multimodal carbon-aware planning are still underserved relative to enterprise control tower demand.** 73 91 42 83
84

2.1

Industry Tier Analysis

Title — European logistics tech competitive tiers

TIER	REPRESENTATIVE PLAYERS	STRATEGIC POSITION	MARGIN AND MOAT IMPLICATION
Global integrators	DSV, DHL, Kuehne+Nagel, Maersk	End-to-end control across forwarding, contract logistics, customs, and customer interfaces.	Strongest cross-sell and data advantages, but software economics are blended with lower-margin physical operations.
Scaled logistics software platforms	Descartes, project44, Transporeon, nShift	Multi-module SaaS with carrier connectivity, workflow depth, and growing ecosystem roles.	Highest potential for software-like margins when embedded in execution and compliance workflows.
Mid-market SaaS specialists	Regional TMS, WMS, last-mile, customs, and planning vendors	Win on implementation speed, vertical fit, and lower total cost of ownership.	Attractive retention where local workflow fit is high, but weaker bargaining power and higher feature replication risk.
Vertical tech and automation vendors	Warehouse robotics, yard, returns, cold chain, parcel, and customs specialists	Own narrow but mission-critical workflows with domain-specific product depth.	Best moats exist where hardware, process redesign, or regulated data models create switching friction.

Source: 78 71 94 96

- Global integrators: DSV, DHL, Kuehne+Nagel, and Maersk are not just logistics operators competing on capacity and procurement. Their annual reports consistently frame digital platforms, data infrastructure, and AI-enabled execution as core to customer retention and wallet share, with DSV highlighting its Enterprise Data Platform and transport management stack, and Maersk positioning digital platforms as the enabler of end-to-end supply chain solutions.⁷²
96
- DSV: The Schenker acquisition materially changes European market structure because it combines one of the most acquisitive global forwarders with a major incumbent network. Management explicitly states the deal made DSV a new global leader and expects integration completion by the end of 2026, which should deepen procurement leverage, customer density, and data scale across forwarding and contract logistics.^{73 70}
- DHL: DHL's strategy remains more portfolio-based than pure forwarding scale, with 2025 acquisitions concentrated in healthcare logistics, returns, and fulfilment. That implies a competitive posture built around vertical solution density and outsourcing capture rather than only transport network breadth, which is strategically relevant in Europe where regulated sectors and reverse logistics complexity support premium pricing.^{71 81}
- Kuehne+Nagel and Maersk: Kuehne+Nagel continues to emphasise digital ecosystem capabilities and AI-supported customs services, while Maersk continues to build integrated logistics and services around platform convenience. Both are using software to increase share of customer supply chain spend, but Kuehne+Nagel appears more focused on service-led orchestration and Maersk on integrated product bundling around its broader transport footprint.⁹¹
94 96
- Scaled software platforms: Descartes and project44 illustrate the strongest software-first competitive models. Descartes continues to describe itself as operating the world's largest collaborative multimodal logistics community and has expanded through acquisitions in TMS and final-mile, while project44 claims a preferred carrier network of 282,000 carriers, over 1.5 billion connected shipments annually, and over 1,000 enterprise brands, indicating meaningful scale in networked visibility and orchestration.^{83 84}
- Mid-market and vertical vendors: This tier remains the broadest part of the European universe and the most likely hunting ground for private equity. The best-positioned companies are not generic point tools, but vendors with strong local compliance content, deep ERP or carrier integrations, or vertical operating specificity in customs, cold chain, returns, warehouse execution, and parcel management, because those attributes reduce direct exposure to platform bundling by integrators and scaled SaaS consolidators.^{83 87 80}

2.2

Sources of Competitive Advantage

- Workflow lock-in: The most durable moat in logistics tech is usually not pure data scale but control of daily execution workflows. DSV's investment in core transport management systems and enterprise data architecture, DHL's embedded contract logistics and healthcare workflows, and Descartes' modular execution stack all show that vendors with booking, customs, routing, audit, or warehouse process ownership sit closer to the transaction and are harder to displace than analytics-only tools.^{72 71 83}
- Network effects: True network effects exist, but only in a limited subset of the stack. Carrier connectivity platforms such as project44 and collaborative logistics networks such as Descartes

benefit when each additional carrier, shipper, or broker improves coverage, data freshness, and onboarding efficiency for the next participant, whereas a standalone WMS or route planning tool usually has customer-specific rather than market-wide network effects.^{84 83}

- Data moats: Data is valuable when it is proprietary, normalised, and linked to action. DSV explicitly describes decades of effort consolidating customer, vendor, employee, and CRM master data into a modern streaming data platform, which is more defensible than raw shipment data alone because it supports quoting, scheduling, exception management, and customer communication across the operating stack.⁷²
- Physical plus digital advantage: Integrators have a hybrid moat that pure SaaS vendors cannot easily replicate. Because DHL, DSV, Kuehne+Nagel, and Maersk control physical execution as well as customer interfaces, they can bundle software into freight, warehousing, customs, and value-added services, making price comparison harder and increasing account stickiness even if the software layer alone would not command a standalone premium.^{81 70 94 96}
- Compliance and customs content: In Europe, one of the least appreciated moats is regulatory workflow depth. Kuehne+Nagel reported noticeable demand for AI-supported customs services in 2025, and vendors that can operationalise customs, trade documentation, emissions accounting, and sector-specific audit trails are likely to sustain better pricing than generic visibility vendors because they sit inside mandatory rather than discretionary workflows.^{91 94}
- AI as moat versus feature: AI only becomes defensible when paired with privileged workflow access and proprietary operating data. project44's acquisition of LunaPath.ai to coordinate agents across multi-step supply chain operations is strategically important because it moves the company from visibility into execution orchestration, while DSV's discussion of autonomous AI agents handling booking, quoting, scheduling, and exception management suggests the same direction among integrators.^{84 72}
- White spaces: The most attractive underserved niches are European SME freight digitisation, multimodal carbon-aware planning, cross-border customs automation for mid-market shippers, and returns or reverse logistics software linked to parcel and warehouse execution. These areas combine fragmented customer bases with rising compliance and service complexity, which creates room for specialist vendors to build sticky products before broader platforms fully absorb the category.^{42 83 91}

2.3

Consolidation Dynamics

- Market structure: Consolidation since 2024 has been driven by two distinct buyer groups. Strategic integrators are buying physical capability and vertical density, while software consolidators are buying adjacent workflow modules and customer bases, which is steadily reducing the number of independent scaled platforms in transport, visibility, and final mile.^{71 89 84}
- DSV and Schenker: This is the single most important competitive event in European logistics structure because it materially increases network density, procurement scale, and enterprise account coverage in forwarding and contract logistics. For software vendors, the implication is mixed: larger integrators can become stronger channel partners or acquirers, but they also become more credible competitors in control tower, visibility, and managed TMS-style offerings.^{70 78}

- DHL and Kuehne+Nagel: DHL's 2025 acquisitions in returns, fulfilment, and specialty pharma, plus Kuehne+Nagel's acquisition of TDN to expand its European groupage network, show that incumbents are still prioritising vertical and network adjacency over large-scale software M&A. That favours vendors serving healthcare, returns, customs, and groupage orchestration because these niches are increasingly strategic to large operators seeking differentiated service bundles.^{71 42 91}
- Software roll-ups: Descartes remains one of the clearest consolidators in logistics software. Its 2025 acquisitions included 3GTMS in transportation management and PackageRoute in final-mile carrier solutions, extending a long-standing strategy of assembling a broad modular platform around multimodal execution, compliance, and ecommerce logistics.^{89 83}
- Visibility to orchestration: project44's acquisition of LunaPath.ai and its July 14, 2026 separation into two businesses, including LSP44 for logistics providers, indicates that the next consolidation phase is not just about adding visibility nodes but about owning agentic workflow infrastructure. That raises the bar for independent visibility vendors, because enterprise buyers increasingly want systems that can recommend and execute actions rather than only report status.^{84 88}
- New entrants and likely consolidators: The most plausible next wave of consolidators are scaled SaaS platforms with strong balance sheets and modular product architectures, plus large integrators seeking compliance, customs, and AI orchestration capabilities. The most vulnerable targets are subscale point solutions in visibility, last mile, and generic TMS layers that lack either proprietary network access or deep regulatory workflow content, while the most resilient independents are likely to be vertical specialists in customs, healthcare logistics software, warehouse execution, and carbon accounting embedded in transport operations.^{83 87 81}

3

Subsector Margin Profiles & Unit Economics

3

Subsector Margin Profiles & Unit Economics

Margin dispersion across European logistics tech is driven less by end-market growth than by revenue architecture. **The highest-quality margin pools sit in execution-critical SaaS and network software with predominantly recurring revenue, while the weakest sit in transaction-led brokerage and hardware-heavy models where gross margin is structurally capped by pass-through costs or field deployment intensity.** ^{115 113 109 84}

For deal teams, the practical implication is that subsector selection matters more than headline logistics digitisation exposure. TMS, WMS, customs, routing, and visibility platforms can sustain software-like economics when embedded in daily workflows, whereas robotics, last-mile operations, and digital freight brokerage require much tighter underwriting around service mix, utilisation, and contract structure to avoid mistaking revenue growth for durable margin. ^{115 108}

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3.1

Margin Benchmarks by Subsector

Title — Margin benchmarks by operating model

SUBSECTOR PROXY	GROSS MARGIN	EBIT OR EBITDA MARGIN	ARR OR RECURRING REVENUE SIGNAL
Logistics SaaS and network software	80% to 87% gross margin.	45% to 53% adjusted EBITDA in scaled leaders.	High recurring mix, typically 90% plus of revenue.
Visibility and control tower platforms	Software-like gross margin inferred when sold as network SaaS.	Lower than mature public SaaS peers, but improving with scale.	project44 reported 34% year over year new ARR growth in Q1 FY27 and multi-year deals drove 67% of new ARR.
Warehouse automation and robotics platforms	Lower blended gross margin than pure SaaS because hardware, installation, and support dilute software economics.	Positive segment EBITDA is achievable at scale, but below pure software.	Ocado states its solutions model is recurring-fee based, with contribution gross margin around 70% relative to a roughly 5% fee.
Digital freight brokerage and transaction-led platforms	Structurally lower gross margin because revenue includes freight pass-through and spot exposure.	EBIT margins are cyclical and volume-sensitive.	ARR is less relevant than shipment volume, take rate, and repeat shipper density.

Source: ^{115 129 113 109 108 84}

- TMS, WMS, customs, and routing software: Public comps indicate the cleanest margin profile in the sector. Descartes generated \$677.2 million of services revenue in fiscal 2026, equal to 93% of total revenue, and reported adjusted EBITDA margin of 45%, while WiseTech reported 87% gross profit margin and 53% EBITDA margin in FY2025, setting a realistic upper bound for scaled logistics software economics. ^{115 129 113}

- Visibility and control tower software: The best available current benchmark is growth quality rather than disclosed profitability, because private leaders do not publish full margin stacks. project44 reported 34% year over year new ARR growth in May 2026, with shipper new ARR up 52% and multi-year deals representing 67% of new ARR, which supports the view that the category can achieve strong recurring economics when it evolves from tracking into workflow orchestration.⁸⁴
- Warehouse automation and robotics: Economics are materially more mixed because software, hardware, maintenance, and deployment services are bundled. Ocado's Technology Solutions revenue reached £561.2 million in FY2025, up 13.0%, and the company states its recurring-fee model can generate about 70% contribution gross margin on a roughly 5% fee, but this remains below pure-play SaaS gross margin once installation, support, and platform R&D are fully considered.^{108 109}
- Digital freight brokerage and last-mile transaction platforms: Reliable Europe-specific public EBIT benchmarks remain sparse, and current primary disclosures do not support a precise cross-company margin table. The underwriting conclusion is still clear: **where revenue is tied to freight bought and sold, gross margin is a take-rate outcome rather than a software outcome, making earnings more cyclical and less durable than in subscription-led execution software.**^{101 115}

3.2

Asset-light vs. Asset-heavy Models

- Asset-light SaaS platforms: The strongest unit economics come from products where implementation is finite, hosting costs are low, and incremental modules carry high contribution margin. Descartes' model is illustrative: 93% of fiscal 2026 revenue came from services, and management says ordinary-course loss of annualised recurring services revenue is typically only 5% to 7%, implying low gross churn and strong lifetime value when customers are embedded in transport, customs, or routing workflows.¹¹⁵
- SaaS retention and contract structure: project44 reported that multi-year deals drove 67% of new ARR in Q1 FY27 and that shipper NRR improved by about 300 basis points quarter over quarter, indicating that contract duration and expansion are improving as visibility products move closer to system-of-action status. For investors, that matters because longer contracts and rising NRR compress effective CAC payback even when headline sales efficiency is not publicly disclosed.⁸⁴
- Hardware and automation models: Asset-heavy businesses absorb lower initial gross margin because they carry bill-of-materials cost, deployment labour, site commissioning, and ongoing field support. Ocado explicitly notes that its model includes funding warehouse automation and then charging recurring fees for use of the platform, which improves lifetime monetisation but lengthens payback and raises execution risk relative to pure software.^{109 107}
- CAC and LTV implications: In asset-light logistics SaaS, CAC is usually front-loaded in enterprise sales and integration, but LTV is supported by low churn, module expansion, and annual price uplifts. In asset-heavy robotics, CAC is effectively blended with solution design and deployment cost, so LTV can still be attractive, but only if utilisation, maintenance attach, and renewal of software and service layers offset the lower upfront margin.^{115 109 3}
- Churn and contract length: The best current disclosed benchmark in logistics software remains low single-digit to high single-digit gross churn for mature platforms and rising multi-year deal mix for growth platforms. By contrast, hardware-heavy models often show lower logo churn

once installed, but that apparent stickiness can mask weaker economics if upgrades, maintenance, and software layers are underpriced relative to the capital intensity of deployment.^{115 84 102}

- Investment conclusion: **Asset-light wins on speed of payback, margin scalability, and resilience through freight cycles; asset-heavy wins only where the vendor controls a full stack of software, maintenance, and workflow redesign that converts hardware from a one-time sale into a long-duration annuity.**^{113 109 3}

3.3

Revenue Quality and Valuation

- Recurring versus transactional mix: Revenue quality is the clearest bridge between operating model and valuation. Descartes derived 93% of fiscal 2026 revenue from services and only 7% from professional services and other revenue, while licence revenue was less than 1%, producing a highly recurring profile that supports premium software multiples and lower earnings volatility.¹¹⁵
- High-quality recurring software: WiseTech's FY2025 disclosures emphasise a recurring revenue model, diversified customer base, 87% gross margin, and 53% EBITDA margin. That combination is what public markets reward most highly in logistics tech because it signals that growth is coming from embedded workflow software rather than freight exposure or project revenue.¹¹³
- Hybrid recurring models: Ocado is a useful middle case for European investors because its solutions business combines recurring platform fees with non-recurring deployment and hardware elements. The company states that recurring revenue is driven by average live modules and that non-recurring revenue also contributed to FY2025 profit flow-through, which means valuation should distinguish between annuity-like software fees and lower-multiple implementation or hardware revenue.^{108 109}
- Transaction-led platforms: Brokerage and spot-market businesses can grow quickly, but valuation support is weaker because revenue is more exposed to freight rates, shipment volumes, and take-rate compression. Even when software is present, investors typically underwrite these models on contribution margin and cycle-adjusted EBITDA rather than on ARR, because the revenue stream is less contractually durable.¹⁰¹
- Valuation implication: **The market consistently pays more for recurring, low-churn, multi-module revenue than for gross merchandise value or hardware-heavy revenue, even inside the same logistics workflow.** In practical terms, a TMS or customs platform with 90% plus recurring revenue, low churn, and expansion potential deserves a structurally higher multiple than a faster-growing brokerage or robotics vendor whose revenue includes pass-through freight, equipment, or project services.^{115 113 109}
- Source limitation note: Current primary public disclosures provide strong benchmarks for listed logistics software and hybrid automation platforms, but not a complete Europe-only set of gross margin, EBIT margin, CAC payback, LTV:CAC, and NRR by subsector. Where private-company metrics were not verifiable from current primary sources, the analysis uses disclosed recurring revenue, churn, contract mix, and margin proxies rather than overstating precision.^{84 115}

4

Sources of Durable Margin

4

Sources of Durable Margin

Durable margin in European logistics tech is created where software becomes part of the operating spine rather than a reporting overlay. The strongest margin pools sit in products that are deeply integrated into ERP, WMS, TMS, customs, and carrier workflows, because replacement risk then includes process redesign, data remapping, partner re-onboarding, and compliance revalidation rather than a simple feature comparison.^{115 132 158}

Across subsectors, **margin durability is highest in execution software and network platforms that combine workflow lock-in with reusable data assets and compliance adjacencies.** By contrast, stand-alone visibility, generic route optimization, and transaction-led brokerage remain more exposed to feature replication and price competition unless they evolve into systems of action with embedded carrier APIs, decision automation, and auditable regulatory outputs.^{159 157 21}

4.1

Switching Cost Architecture

- The core retention mechanism in logistics tech is integration depth. When a platform sits between ERP, WMS, TMS, carrier APIs, customs filing, and document generation, the customer is not only buying software but also a working data model for orders, inventory, rates, milestones, invoices, and compliance events, which materially raises the cost of replacement.^{163 158 132}
- Descartes is a useful public benchmark because its products span routing, shipment execution, customs, trade data, audit and pay, and final mile, all delivered through a collaborative logistics network. That breadth helps explain why the company reported that ordinary-course loss of annualized recurring services revenue is typically only 5% to 7%, a strong proxy for low gross churn once customers are embedded in multiple workflows.¹¹⁵
- Data migration is a real moat because logistics master data is unusually messy. Customer-specific carrier contracts, lane logic, SKU and packaging hierarchies, customs attributes, appointment rules, and exception workflows are often encoded across several systems, so a vendor change requires not just data export but process re-implementation and partner testing. This is consistent with OpenText's positioning that ERP modernization and supply-chain reconnection require pre-built adapters and proven methodologies to reduce go-live risk.^{163 158}
- Academic work on logistics digital twins reinforces the same point from a technical angle. A 2023 review notes that digital twins for logistics and supply chain systems require integration across heterogeneous data sources and advanced analytics layers, implying that once a platform becomes the operational representation of the network, switching means rebuilding both data interoperability and decision logic.¹⁴⁸
- Implementation complexity is also a segmentation filter. Survey evidence on RTLS adoption found that integration difficulty, infrastructure requirements, and calibration effort are major barriers for SMEs, which implies that vendors able to package low-friction deployment for the mid-market can create stickiness without enterprise-scale services intensity.¹⁴⁵
- Investment implication: **WMS, TMS, customs, and warehouse execution platforms have the most defensible switching-cost architecture when they own daily transactions and exception**

handling, not just dashboards. The margin benefit is twofold: lower churn and higher expansion efficiency, because adjacent modules can be sold into an already-integrated data and workflow estate.^{115 132}

4.2

Data and Network Advantage

Title — Margin moats by logistics tech model

SUBSECTOR MODEL	PRIMARY MOAT TYPE	WHY IT SUPPORTS MARGIN DURABILITY	MARGIN DURABILITY VIEW
TMS WMS customs execution SaaS	Workflow lock in plus proprietary operational data	Embedded in daily execution, low churn, strong module upsell.	Highest.
Freight visibility and control tower platforms	Carrier network density plus data graph quality	Better coverage and ETA accuracy improve with each connected carrier and shipment.	High if tied to actioning workflows.
AI planning and orchestration tools	Model performance plus privileged context and execution rights	Value rises when AI can act on live constraints, contracts, and exceptions.	Medium to high.
Digital freight brokerage	Liquidity and pricing data	Useful data advantage, but take rates remain exposed to cycle and competition.	Medium to low.

Source: ^{159 157 132 144}

- True network effects are concentrated in a narrow part of the stack. Visibility and collaborative execution platforms improve as more carriers, shippers, and logistics providers connect, because each new participant expands lane coverage, event density, and onboarding reuse for the next participant.^{159 157}
- project44 is currently the clearest private example of this moat. As of April 8, 2026, it stated that its preferred carrier network covered 282,000 carriers, connected more than 1.5 billion shipments annually, and served over 1,000 leading brands; in July 2026 it further described LSP44 as API infrastructure already used by 9 of the world's 10 largest logistics providers. Those figures matter because they indicate a scale threshold where data freshness, carrier connectivity, and workflow standardization become difficult for smaller rivals to replicate.^{159 157}
- Descartes represents a different but equally durable network model. Its filings describe a large collaborative multimodal logistics community and emphasize network-based solutions for customs, security filings, routing, and transport execution, which means the moat is not only data volume but also established digital relationships among shippers, LSPs, brokers, and government agencies.^{132 115}
- The key distinction for investors is between data exhaust and actionable data advantage. Academic work on supply-chain visibility using graph neural networks and federated learning argues that visibility is constrained by privacy, security, and cross-border data-sharing limits, so vendors that can normalize fragmented data and train models without requiring raw-data centralization may gain a structural edge in Europe's regulated environment.¹⁴⁹

- AI planning tools only become a moat when they sit on top of proprietary context and execution permissions. project44's 2026 agent portfolio explicitly frames the foundation as a decade of context, skills, and orchestration, while recent arXiv work on hyperconnected LTL planning and dynamic routing shows that optimization quality improves when systems can access real-time network conditions, consolidation opportunities, and multi-party constraints.^{159 144 146}
- Investment implication: **freight visibility has durable margin only when it graduates into orchestration, carrier API infrastructure, or decision automation.** Pure tracking data is increasingly commoditizable, but a network that improves ETA quality, exception resolution, procurement decisions, and automated execution can sustain better pricing and lower churn because the customer is buying outcomes rather than telemetry.^{159 87}

4.3

Value-Added Services as Margin Enhancers

- Value-added services are the cleanest route from good software margins to durable software margins because they monetize the same workflow twice. Once a vendor already captures shipment, inventory, or customs data, adjacent modules such as emissions accounting, denied-party screening, trade content, freight audit, document generation, and exception automation can be sold at very high incremental gross margin.^{132 115}
- Compliance-by-design is especially attractive in Europe because regulation is creating standardized data requirements. Regulation (EU) 2026/1030, which entered into force on June 1, 2026, establishes a common methodology aligned with EN ISO 14083:2023 for greenhouse-gas emissions accounting of transport services, creating a direct monetization path for vendors that can embed auditable emissions calculation and reporting into TMS, visibility, and procurement workflows.^{21 160}
- The margin logic is strong because sustainability reporting is largely software and data processing, not physical execution. For an incumbent TMS or visibility vendor, adding carbon calculation, lane-level emissions benchmarking, and optimization recommendations should carry materially higher incremental margin than acquiring a new core execution customer, since the data ingestion and user workflow are already in place. This is an inference from the regulation's standardized methodology and from the modular SaaS economics disclosed by logistics software leaders.^{21 115}
- Customs and trade compliance remain another high-quality margin enhancer. Descartes' filings highlight customs declarations, security filings, sanctioned-party screening, tariff and duty calculation, and regulatory content as core solutions, and these modules are structurally sticky because they sit inside mandatory cross-border workflows rather than discretionary analytics budgets.¹³²
- Agentic AI is the next likely upsell layer, but only for vendors with sufficient workflow authority. project44's 2026 product positioning around AI agents for shipper operations suggests a monetization path from visibility into premium automation, where the customer pays not just for insight but for exception handling, decision support, and task execution across transport workflows.¹⁵⁹
- The most defensible subsectors on this basis are TMS and customs platforms, WMS with embedded labor and automation orchestration, and visibility networks that have already crossed into system-of-action territory. **These categories combine low churn, high attach potential for**

compliance and sustainability modules, and a credible path to AI upsell, which is the strongest formula for sustained gross margin and expanding EBITDA over time. ^{115 87 160}

5

AI & Technology Industrialisation

5

AI & Technology Industrialisation

AI in European logistics tech is moving from experimentation to workflow industrialisation, but adoption remains uneven by use case and by company scale. The strongest evidence of durable value sits in execution-adjacent applications such as spot pricing, customs document handling, exception management, warehouse orchestration, and network decisioning, where AI is attached to proprietary operating data and can change throughput, service levels, or labor intensity rather than merely improve reporting.^{189 178 71}

For investors, the key distinction is not whether a vendor has AI features, but whether AI is embedded deeply enough to alter unit economics and retention. **AI is a moat only when it is coupled to workflow authority, proprietary context, and a reusable data layer; otherwise it is increasingly a table-stakes feature that will compress differentiation over time**^{84 180 148}

5.1

AI Adoption in Logistics Tech

AI adoption is now broad at the category level, but still shallow in many deployments. McKinsey's latest logistics survey of more than 260 respondents found high digital adoption, with 54% of large shippers already having implemented at least five digital use cases and 59% expecting to implement ten or more within three years; 55% of large enterprise respondents had already implemented at least two gen AI use cases.¹⁷⁸

The practical adoption frontier is concentrated in planning and execution support rather than full autonomy. BCG and Alpega's January 2026 survey of more than 180 logistics experts across Europe, North America, Asia, and the Middle East frames current usage around transport planning, predictive demand analytics, and operational support, indicating that AI has become a top management agenda item but still requires significant scaling work to deliver impact at enterprise level.¹⁸⁹

- Planning and pricing: Kuehne+Nagel states that its Air Logistics unit uses AI for spot-pricing decisions, reducing manual market and capacity analysis and accelerating customer response times through its myKN platform. This is one of the clearest examples of AI moving from analytics into revenue-critical quoting workflows.¹⁸⁰
- Demand forecasting and scenario support: McKinsey identifies demand forecasting and capacity planning among the 28 logistics digital use cases already being deployed, while BCG highlights predictive analytics as a core current application. The implication is that forecasting is no longer an emerging use case, but it is not yet a strong moat because models are increasingly available through standard cloud and application layers.^{178 189}
- Freight audit, customs, and document workflows: Kuehne+Nagel's AI assistant Claire is used to summarise complex customs documents and prepare declarations faster, while DHL says robotics and AI are improving quality and efficiency in customer service, customs clearance, fulfillment, and service logistics. These are attractive use cases because they sit inside mandatory workflows and can convert labor-heavy processes into scalable software margin.^{180 71}
- Warehouse orchestration: Recent arXiv work on reinforcement learning for SAP Logistics Execution modeled warehouse task allocation, inventory movement, and order picking on a

synthetic dataset of 300,000 transactions and reported 60% lower processing times versus traditional methods. This is not a market benchmark, but it is useful technical evidence that warehouse AI is moving beyond static rules into deployable orchestration logic.¹

The adoption pattern matters for margin analysis. **AI is already material in logistics where it removes manual touches from quoting, customs, and exception handling, but route optimization, forecasting, and generic copilots are becoming standard product expectations rather than standalone sources of pricing power** ^{180 178}

5.2

Comparative Advantage of AI Deployments

The clearest advantage of AI early adoption is not headline revenue growth but faster operating leverage in labor-intensive workflows. DHL explicitly links digitalization, automation, and AI to improved quality and efficiency across customer service, customs clearance, fulfillment, and service logistics, while Kuehne+Nagel says it has moved from initial AI development into a scaling phase with agents supporting order intake, customer communication, document processing, and exception handling globally.^{71 180}

Title — AI deployment advantage by logistics model

MODEL	EARLY ADOPTER EFFECT	MARGIN IMPLICATION
Integrated logistics operators	Faster document handling, quoting, and exception resolution across existing volume base.	Better labor productivity and service consistency, but gains are partly blended into broader operating margins.
Network software platforms	AI improves retention, upsell, and workflow depth when attached to carrier and shipment data.	Highest incremental software margin if AI drives NRR and multi-year contract expansion.
Mid-market point solutions	Faster feature parity through embedded AI APIs, but weaker proprietary context.	Limited durable uplift unless tied to unique data or regulated workflows.
Robotics and automation vendors	AI can improve throughput and utilization, but economics remain constrained by hardware and deployment intensity.	Margin uplift is real but diluted by capex, installation, and support costs.

Source: ^{84 71 180 1}

project44 provides the strongest current commercial signal that AI-led workflow expansion can translate into software economics. In Q1 FY27, announced on May 18, 2026, the company reported 34% year over year growth in new ARR, 52% shipper new ARR growth, 67% of new ARR from multi-year deals, and an approximately 300 basis point quarter over quarter improvement in shipper NRR, which management explicitly tied to AI agent momentum and deeper operational embedding.⁸⁴

By contrast, laggard implementations tend to bolt AI onto fragmented workflows without redesigning the operating model. McKinsey notes that the logistics technology landscape remains highly fragmented and that companies still need a clear view of what processes and technology should do which activities, implying that many deployments remain additive rather than transformative.¹⁷⁸

- Financial divergence: Early adopters can convert AI into higher NRR, longer contract duration, and lower service labor per transaction, which is the most attractive path to EBITDA expansion in software-led models. Laggards may still incur model, integration, and change-management cost without achieving measurable retention or pricing benefits.^{84 178}
- Operational divergence: Kuehne+Nagel's stated goal is to compress processes that previously took hours into near-instant workflows, while DHL emphasizes quality and efficiency gains in operational functions already at scale. That suggests the biggest advantage accrues to incumbents with enough transaction density to amortize model development across large workflow volumes.^{180 71}
- Competitive divergence: Early adopters build proprietary feedback loops because every quote, declaration, exception, and shipment outcome improves future model performance. Laggards can often buy similar base models, but they cannot quickly replicate years of labeled operational context and embedded user behavior. This is an inference supported by the workflow depth described by Kuehne+Nagel and project44.^{180 84}

The investment conclusion is that **AI early adoption creates durable advantage only when it compounds existing scale advantages in data, workflow ownership, and customer integration**. In pure feature markets, the lead time between innovator and follower is likely too short to sustain excess margin for long.^{178 189}

5.3

Evaluating Emerging Technologies

The most important platform bets in logistics are digital twins, autonomous warehouse robotics, and agentic orchestration layers. Their durability depends less on technical novelty than on whether they become part of the operating system for planning and execution, with persistent data ingestion, simulation, and closed-loop actioning.^{148 173}

Digital twins remain strategically attractive but commercially uneven. The 2023 arXiv review on logistics and supply chain digital twins argues that they can improve decision-making, transparency, and resilience, but also highlights practical challenges around heterogeneous data integration, trustworthiness, and implementation complexity, which means many deployments will remain expensive overlays unless tied directly to execution systems.¹⁴⁸

Autonomous warehouse systems are further along the industrialisation curve than full supply-chain twins. DHL's 2025 report states that robotics and AI are already contributing to improved quality and efficiency, and the 2026 AI and ML roadmap for smart manufacturing identifies autonomous systems, digital twins, robotics, and supply chain optimization as active industrial application domains rather than purely experimental ones.^{71 173}

Agentic AI procurement and control-tower orchestration are the highest-upside but also the most overhyped category. project44's May 2026 results indicate that AI agents are already helping drive ARR growth and deeper customer embedding, but the moat exists only where agents can act on a proprietary network, trusted data graph, and execution permissions rather than simply generate recommendations.⁸⁴

- Digital twins: Long-term viable in network design, resilience planning, and warehouse simulation, but not unique by default. **They become defensible only when fed by live operational data and linked to execution rights inside TMS, WMS, or control towers**¹⁴⁸

- Autonomous warehouse robotics: More durable than generic AI copilots because deployment requires software, physical integration, safety validation, and process redesign. Even so, the moat is strongest in the orchestration software and installed base, not in commodity robot hardware alone.^{71 1}
- Agentic procurement and control towers: Potentially high value where platforms can automate tendering, exception handling, and re-planning across many carriers and modes. However, uniqueness is limited if the vendor lacks proprietary carrier connectivity, historical rate context, and customer workflow authority.^{84 180}
- Commodity feature risk: Forecasting assistants, generic route optimization, and document copilots are increasingly easy to replicate through foundation models and standard software tooling. These features may improve win rates, but they are unlikely to sustain premium pricing unless bundled into a broader execution platform.^{189 178}

For investors, the durable technology thesis is narrow. **Back platforms where AI is inseparable from execution, compliance, or installed automation, and discount vendors whose AI story is primarily interface-level productivity without proprietary data, workflow control, or measurable retention uplift**^{84 71 148}

6

Regulatory Tailwinds & Risk Vectors

6

Regulatory Tailwinds & Risk Vectors

European logistics tech is entering a phase where regulation is no longer a background constraint but a direct driver of product design, pricing power, and customer urgency. **The investable opportunity is concentrated in software layers that convert mandatory emissions, data-sharing, and security obligations into embedded workflows, while the main risk sits with vendors exposed to volatile transport costs, fragmented cross-border rules, or weak control over data governance.** ^{203 24 31}

For investors, this section sharpens the distinction between compliance-adjacent platforms and discretionary tools. **Regulation is raising the floor on demand for emissions accounting, auditable data exchange, and resilient network planning, but it is also increasing execution risk for subscale vendors that lack verification capability, secure data architecture, or the balance sheet to absorb implementation complexity.** ^{207 204 229}

6.1

Cost and Compliance Opportunities

The EU ETS extension to maritime has applied since 1 January 2024, with a phase-in that requires shipping companies to surrender allowances for 40% of verified emissions for 2024, 70% for 2025, and 100% from 2026 onward; methane and nitrous oxide are also included from 2026. That timetable matters for logistics tech because 2026 is the first year in which the full carbon-cost signal is visible in procurement, routing, and customer invoicing workflows rather than being partially deferred. ^{202 203 217}

CountEmissionsEU, Regulation (EU) 2026/1030, entered into force on 1 June 2026 and establishes a common methodology for greenhouse-gas emissions accounting of transport services aligned with EN ISO 14083:2023. Although the regime standardises calculation rather than imposing a universal carbon tax, it creates a common data model for lane-level emissions reporting, which is exactly the kind of regulatory standard that supports premium attach rates for TMS, visibility, audit, and procurement platforms. ^{21 207}

Title — EU carbon compliance milestones affecting logistics tech

MEASURE	EFFECTIVE DATE	OPERATIONAL IMPLICATION	TECH OPPORTUNITY
EU ETS maritime inclusion	1 January 2024	Carbon cost begins to flow into maritime transport economics.	Carbon-cost visibility in booking, procurement, and customer billing.
EU ETS phase-in at 70%	Calendar year 2025	Partial but material cost pass-through pressure on shippers and forwarders.	Scenario tools for surcharge design and contract repricing.
EU ETS phase-in at 100% plus CH4 and N2O coverage	Calendar year 2026	Fuller compliance burden and broader emissions scope.	Verified emissions engines and audit-grade reporting.

MEASURE	EFFECTIVE DATE	OPERATIONAL IMPLICATION	TECH OPPORTUNITY
CountEmissionsEU in force	1 June 2026	Standardised methodology for transport emissions calculation.	Embedded emissions accounting, benchmarking, and optimisation modules.
CBAM definitive regime	1 January 2026	Importers face carbon reporting and payment obligations on covered goods.	Trade, customs, and landed-cost software linking product and transport data.

Source: 202 203 21 204

- Cost pass-through: Pass-through is most feasible in enterprise contracts where carriers, forwarders, and 3PLs already use indexed fuel and surcharge mechanisms. It is materially harder in fragmented SME road freight and spot-heavy forwarding, where vendors that can automate carbon allocation, surcharge logic, and customer-facing audit trails should gain pricing relevance.^{202 71}
- Compliance stack opportunity: The highest-value products are not stand-alone carbon dashboards but workflow modules that connect shipment data, carrier factors, contract terms, and invoice logic. **This favors TMS, visibility, customs, and freight-audit vendors that can turn emissions accounting into a system-of-record function rather than a reporting add-on.** ^{21 210}
- CBAM adjacency: The CBAM definitive regime applies from 1 January 2026 for covered imported goods, and the Commission has explicitly assessed whether CBAM could be extended to transport emissions and transportation services, concluding that measurement and verification across multiple logistics operators would be complex. That complexity is strategically positive for software vendors because it increases the value of interoperable trade, customs, and emissions data layers even before any broader transport extension is adopted.^{204 210}
- Investment implication: **Regulation is creating a monetisable compliance layer with software-like economics.** The best-positioned vendors are those able to sell emissions calculation, verification support, carbon-aware procurement, and customer reporting as high-margin modules into already embedded transport workflows.^{21 203}

6.2

Data Sovereignty and Security

The EU Data Act was published on 22 December 2023 and has applied since 12 September 2025. Its core effect is to make data generated by connected products and related services more accessible to users and, under defined conditions, to third parties, while also introducing safeguards on switching between cloud and edge providers and protections against unlawful third-country government access to non-personal data.²⁴

For logistics tech, the commercial significance is substantial because fleets, trailers, warehouse equipment, telematics devices, and industrial IoT systems generate operational data that has historically been difficult for shippers or 3PLs to extract from OEM-controlled environments. **The Data Act therefore cuts both ways: it can weaken proprietary data hoarding by hardware or platform incumbents, but it also rewards vendors that can operationalise lawful access, interoperability, and secure cross-party sharing better than smaller rivals.** ²⁴

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- **Competitive moat:** Vendors with strong consent management, audit logs, API governance, and interoperable data models can turn compliance into a moat because customers increasingly need proof that operational data can be shared, ported, and governed without breaking service continuity. In logistics, that favors platforms already sitting between telematics, ERP, WMS, TMS, and carrier networks, because they can normalize fragmented machine and shipment data into a reusable control layer.^{24 205}
- **Structural liability:** The same rules can erode margins for vendors whose differentiation depends on exclusive access to device-generated data or on high switching friction in cloud environments. If a platform lacks superior analytics, workflow depth, or compliance tooling, broader data portability can compress lock-in and shift bargaining power toward customers and integrators.²⁴
- **Security burden:** The Data Act does not relax security expectations; it raises them by requiring controlled sharing and safeguards around international access requests. That increases the cost base for encryption, access control, data lineage, contractual governance, and regional hosting architecture, which is manageable for scaled vendors but can be margin-dilutive for subscale point solutions.^{24 206}
- **Investor read-through: **Data sovereignty is becoming a selection criterion, not a hygiene factor.**** The winners are likely to be platforms that combine interoperability with trusted governance, because they can market compliance and portability as customer benefits while still defending margin through workflow ownership, analytics quality, and multi-party network integration.^{24 212 205}

6.3

Macro and Geopolitical Risks

Macro risk remains meaningful even as regulation creates demand. The European Commission's Spring 2025 forecast projected EU real GDP growth of 1.1% in 2025 and 1.5% in 2026, but it explicitly highlighted heightened global policy uncertainty and trade tensions, while a separate Commission survey found that geopolitical tensions and trade fragmentation are already pushing firms toward digitisation, monitoring, and supply-chain redesign.^{228 31}

Energy remains a direct sensitivity for logistics operations and for warehouse-intensive tech models. Eurostat reported that average EU non-household electricity prices were €19.03 per 100 kWh in the first half of 2025 and €18.37 in the second half, down 3.5%, but dispersion across member states remained wide, which means warehouse automation, cold-chain, and compute-heavy operations still face uneven regional cost exposure.^{224 222}

DHL's 2025 Annual Report states that demand in the European road freight market remained subdued in 2025 while costs stayed high, reflecting the difficult combination of soft volumes and persistent operating inflation. For logistics tech vendors, that backdrop is double-edged: it supports ROI-led software demand, but it can also elongate sales cycles, pressure SME budgets, and reduce transaction-linked revenue in brokerage or usage-based models.^{71 226}

- **Freight cycle exposure:** Spot-heavy brokerage, last-mile, and usage-priced visibility tools remain most exposed to weak freight demand and rate compression. By contrast, compliance, customs, WMS, and execution software with contracted recurring revenue should prove more resilient through a soft transport cycle.^{71 31}
- **Geopolitical disruption:** The Commission's economic security and foresight work points to weaponisation of dependencies, tariffs, export controls, and critical supply-chain vulnerabilities

as live risks rather than abstract scenarios. That supports demand for control towers, supplier mapping, customs automation, and scenario-planning tools, especially where customers are redesigning networks around nearshoring or dual sourcing.^{227 229}

- Strategic mitigation: Investors should prefer vendors with low direct freight-rate exposure, multi-country hosting and compliance architecture, and products tied to mandatory workflows such as customs, emissions, or warehouse execution. Portfolio companies should also build carbon-cost scenario engines, regional data-residency options, and supplier-risk monitoring into the core roadmap rather than treat them as optional modules.^{24 21 227}
- Bottom line: **The macro and geopolitical environment increases volatility, but it also strengthens the case for mission-critical logistics software.** The most durable margin profiles should continue to sit with vendors that help customers absorb regulatory complexity and network disruption without taking principal risk on freight, fuel, or hardware-heavy deployment.³¹

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M&A Activity & Valuation Benchmarks

7

M&A Activity & Valuation Benchmarks

European logistics tech M&A from 2024 to August 25, 2026 shows a clear buyer preference for assets that deepen workflow ownership, add regulated or high-service logistics capabilities, or convert visibility into execution. **The market is rewarding platforms that combine recurring software revenue with network density, compliance content, or automation adjacency, while more transactional and hardware-heavy models continue to clear at lower and more deal-specific multiples.** ^{264 258 262}

The strategic direction is consistent with the earlier margin analysis in this report. Buyers are paying up for TMS, customs, control-tower, and final-mile software where attach rates, retention, and cross-sell are strongest, and they are using M&A to close product gaps faster than internal build cycles would allow. ^{252 249 250}

7.1

Recent M&A Transactions

- The most consequential European logistics transaction in the period was DSV's completion of the Schenker acquisition, announced with an enterprise value of approximately DKK 106.7 billion, or about EUR 14.3 billion. Strategically, the deal is about network density, procurement leverage, and end-to-end control rather than pure software, but it materially raises the competitive bar for independent control-tower, managed TMS, and orchestration vendors selling into large European shippers. ²⁷²
- Kuehne+Nagel's acquisition of Spanish road logistics provider TDN in 2025 was smaller in headline size but highly relevant in strategic logic. TDN handled more than one million shipments in 2024, and Kuehne+Nagel explicitly framed the deal around denser Iberian groupage coverage, better lead times, and higher departure frequency, which reinforces the thesis that incumbents are buying execution density in lanes where software, service quality, and network economics reinforce each other. ²⁵⁷
- DHL's 2025 acquisition program was notably capability-led rather than scale-led. Its Q4 2025 investor presentation identifies Inmar Supply Chain Solutions as a returns specialist including software, processing, and re-marketing, alongside CRYOPDP in temperature-controlled pharma logistics and GT in specialized final-mile healthcare delivery, indicating a deliberate push into higher-value, workflow-rich niches where technology and service are tightly coupled. ²⁵⁰
- Descartes remained the clearest software consolidator in the period. It acquired 3GTMS in March 2025 to deepen transportation management and carrier connectivity, and its fiscal 2026 filings show it also acquired PackageRoute on June 18, 2025 for approximately \$1.9 million net of cash, extending its final-mile carrier solution footprint. ^{252 256}
- project44's April 9, 2026 acquisition of LunaPath.ai is strategically important because it marks the category shift from visibility to agentic orchestration. Management stated that LunaPath's agents would be combined with project44's logistics data graph to move from understanding what happened to determining what action should be taken, which is exactly the kind of workflow deepening that supports premium retention and valuation. ²⁴⁹
- Across the broader market, Deloitte reports that European logistics M&A momentum in 2024 was dominated by smaller and mid-sized deals, with haulage and transport representing 29% of

deals and freight forwarding and supply chain solutions 15%. The implication for deal teams is that the market remains highly active below mega-cap scale, which supports continued sponsor-backed platform building and bolt-on execution in fragmented software and service niches.²⁶⁴

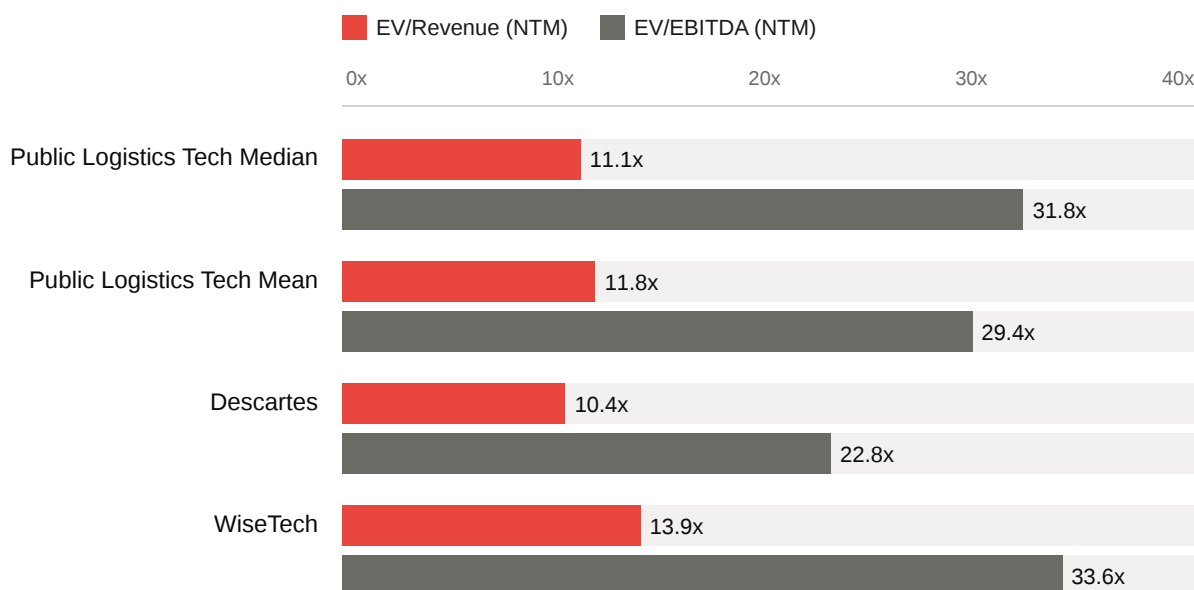
7.2

Valuation Trends and Multiples

Title — Logistics tech valuation benchmarks

SEGMENT OR COMP SET	EV/REVENUE	EV/EBITDA	READ-THROUGH
Public logistics tech median, Sep 2025	11.1x NTM	31.8x NTM	Premium for recurring, workflow-embedded software.
Public logistics tech mean, Sep 2025	11.8x NTM	29.4x NTM	Broadly supports double-digit revenue multiples for scaled platforms.
Descartes, Sep 2025	10.4x NTM	22.8x NTM	Benchmark for profitable logistics execution software.
WiseTech, Sep 2025	13.9x NTM	33.6x NTM	Upper-end multiple for global network software with strong margins.

Source: ²⁵⁸



- Public market benchmarks indicate that scaled logistics software continued to trade at premium levels through late 2025. Capstone's September 30, 2025 comp set shows a median 11.1x NTM EV/Revenue and 31.8x NTM EV/EBITDA, with Descartes at 10.4x and 22.8x and WiseTech at 13.9x and 33.6x, which is consistent with the market paying most for recurring revenue, global network effects, and high EBITDA conversion.²⁵⁸
- Drake Star's September 22, 2025 software peer work points in the same direction, with a median 9.1x EV/2025E revenue and 31.0x EV/2025E EBITDA across its supply chain software peer group. Within that set, WiseTech screened at 20.0x revenue and Descartes at 25.6x EBITDA,

reinforcing that the top end of the range is reserved for category leaders with superior growth, margin, and network positioning.²⁶²

- The multiple spread between platform assets and bolt-ons remains wide. Platform-grade assets with multi-module SaaS, low churn, and international expansion potential can justify low-teens EV/Revenue and 20x to 30x plus EBITDA ranges in public comps, while bolt-ons are more often priced off strategic fit, customer overlap, and cross-sell potential rather than headline market multiples. This is an inference from the public comp ranges and the acquisition behavior of Descartes, DHL, and project44.^{258 252 249}
- Asset-heavy and service-heavy logistics deals continue to clear at lower EBITDA multiples than pure software. MCF Corporate Finance highlights UPS's acquisition of Frigo-Trans at an exceptional 14.5x EV/EBITDA in 2024, which is notable precisely because it is exceptional for a logistics services asset rather than normal for the category.²⁵⁵
- For underwriting purposes, the most useful benchmark split is by revenue architecture. **Recurring execution software, customs, TMS, WMS, and network orchestration deserve software-style multiples; final-mile, brokerage, and automation assets require a discount unless software content, maintenance annuities, or installed-base lock-in are strong enough to normalize earnings through the freight cycle.**^{258 262 264}
- A source limitation is worth stating explicitly. Reliable, current Europe-only private transaction multiple datasets for TMS, WMS, visibility, last-mile software, and warehouse robotics are not comprehensively disclosed in public primary sources, so the benchmarks above rely on current public trading comps and disclosed transaction examples rather than a fabricated private-market precision that cannot be verified.^{258 262}

7.3

Strategic Investment Theses

- The dominant private equity thesis is still platform-plus-bolt-on consolidation in fragmented workflow software. The logic is strongest in TMS, customs, final-mile, and control-tower adjacencies where a sponsor can buy a sticky core product, add modules or geographies through M&A, and expand EBITDA through cross-sell rather than pure cost takeout.^{264 252}
- A second thesis is vertical specialization around regulated or service-critical niches. DHL's focus on returns software and processing, temperature-controlled pharma logistics, and specialized healthcare final mile shows where strategics see premium value: categories where compliance, service reliability, and workflow complexity support better pricing and lower churn than generic freight execution.²⁵⁰
- A third thesis is that visibility alone is no longer enough, but visibility plus action can still command premium growth capital. project44's LunaPath acquisition and subsequent AI-led positioning suggest that growth investors are now underwriting orchestration, agent infrastructure, and decision automation rather than stand-alone milestone tracking, because those layers have a clearer path to NRR expansion and system-of-action status.^{249 84}
- Warehouse automation remains attractive, but the investment thesis is more selective than in pure software. The best opportunities are in orchestration software, installed-base service revenue, and hybrid automation platforms rather than in undifferentiated hardware, because margin durability depends on software control and maintenance annuities more than on robot

unit sales. This inference is consistent with the broader valuation discount applied to asset-heavy models relative to software peers.^{258 3}

- The subsectors attracting the clearest premium valuations are global logistics software platforms, customs and compliance software, and orchestration layers with strong carrier or shipment network effects. Public comps show that investors continue to pay most for businesses that combine high recurring revenue with embedded workflow authority, while strategic deal flow shows that buyers are willing to pay for returns, healthcare, and final-mile niches when those niches add defensible service and data depth.^{258 250}
- For European deal teams, the highest-conviction thesis remains narrow. **Back platforms that own execution workflows, can attach compliance and AI modules, and have room for bolt-on consolidation; be more cautious on transaction-led brokerage, generic visibility, and hardware-first models unless there is clear evidence of recurring software monetization and durable installed-base lock-in.**^{262 264}

8

Customer Economics & Demand-Side Dynamics

8

Customer Economics & Demand-Side Dynamics

Customer demand in European logistics tech is bifurcating between large shippers that want fewer strategic platforms and smaller operators that still buy tactically around immediate ROI.

For investors, the highest-quality revenue sits with vendors that either become a consolidated control layer for enterprise procurement or deliver low-friction, fast-payback modules for the SME base, while the weakest position remains in undifferentiated point tools caught between those two buying motions. ^{282 278 284}

A second demand-side shift is the rising role of outsourced logistics partners as software gatekeepers. As transportation and warehousing are increasingly delegated to 3PLs and, selectively, 4PL-style orchestrators, platform choice is moving upstream from the shipper IT team toward the operating partner that controls execution, data capture, and carrier connectivity. **That dynamic increases retention for embedded platforms, but it also concentrates customer power in fewer channel relationships and raises dependency risk for vendors that do not own a direct system-of-action position.** ^{293 284 273}

8.1

Procurement Behavior Analysis

European shipper procurement is increasingly split between suite rationalisation at the core and selective best-of-breed at the edge. Gartner's Europe-focused 3PL research indicates that logistics leaders are actively reassessing provider portfolios for outsourced European operations, while market commentary around TMS consolidation points to shrinking vendor choice and greater emphasis on acquisition-resistant platforms with broad workflow coverage rather than isolated tools. ^{278 283}

- Consolidation logic: Enterprise shippers increasingly want a smaller number of strategic vendors spanning transport execution, visibility, compliance, and analytics because integration cost and governance burden now matter almost as much as feature depth. This is consistent with broader procurement-tech evidence that many organisations remain dissatisfied with fragmented stacks and are evaluating consolidation to reduce operating complexity. ^{287 289}
- Best-of-breed persistence: Best-of-breed remains viable where the workflow is regulated, country-specific, or commercially differentiated, especially in customs, carbon accounting, parcel management, and niche carrier connectivity. In practice, buyers are not choosing pure standardisation or pure fragmentation, but a hub-and-spoke model with one core platform and a limited set of specialist modules connected by APIs. ^{281 289}
- Buying criteria shift: Procurement teams are placing more weight on implementation risk, interoperability, and vendor durability because consolidation in TMS and adjacent categories raises the risk of roadmap disruption after acquisition. That favours scaled vendors with proven ERP and carrier integrations, but it also creates room for specialist software that can plug into incumbent suites without forcing a full replacement. ^{283 280}
- Economic implication for vendors: Consolidation buyers usually produce larger ACVs, longer sales cycles, and stronger multi-module retention, while best-of-breed buyers produce faster land motions but higher replacement risk unless the product owns a mandatory workflow. **The most attractive vendors are those that can sell as either the strategic core or the indispensable specialist, but not those positioned as a nice-to-have overlay.** ^{283 278 289}

Title — Shipper procurement patterns and vendor implications

PROCUREMENT PATTERN	TYPICAL BUYER	VENDOR IMPLICATION	MARGIN QUALITY VIEW
Core platform consolidation	Large multinational shippers and lead 3PLs	Higher ACV, deeper integration, lower churn, longer sales cycle.	High
Hub plus specialist modules	Upper mid-market shippers	Good expansion potential if APIs and compliance modules are strong.	Medium to high
Tactical best-of-breed point buying	SMEs and local operators	Faster wins but lower contract durability and higher price sensitivity.	Medium to low
Outsourcer-led platform selection	Shippers delegating execution to 3PL or 4PL partners	Strong indirect distribution, but vendor risks channel concentration.	High if embedded

Source: 278 289 283 287

8.2

Enterprise vs. SME Economics

The enterprise versus SME split is one of the clearest determinants of margin quality in European logistics tech. Eurostat's 2025 EU data show that ERP usage reached 88.71% among large enterprises, 69.93% among medium enterprises, and only 41.08% among small enterprises, while 53% of all EU enterprises used specialised e-business software in 2025, confirming that digital maturity and integration readiness still rise sharply with company size.²⁹¹
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- Enterprise economics: Large shippers support higher contract values because they need multi-country deployment, carrier connectivity, auditability, and integration into ERP, WMS, and procurement systems. The trade-off is heavier pre-sales, longer implementation, and more bespoke service load, but once embedded these accounts usually deliver better retention and broader module attach.^{291 278}
- SME economics: SMEs are more constrained by upfront cost, installation effort, and internal capability. A 2025 arXiv survey on RTLS adoption found that respondents identified upfront cost, installation effort, and integration difficulty as dominant barriers, with most indicating acceptable upfront investments below \$10,000 and preferring low-infrastructure deployments.¹⁴⁵
- Product design implication: SME vendors win when they minimise services intensity through self-serve onboarding, prebuilt connectors, and clear payback tied to labor, routing, or compliance savings. Enterprise vendors win when they can justify complexity with measurable network savings, governance benefits, and cross-functional workflow control.^{145 289}
- Margin implication: Enterprise customers are usually more valuable on LTV despite higher CAC because churn is lower and expansion paths are wider, especially for compliance, analytics, and AI modules. SMEs can still be attractive where distribution is partner-led or product-led, but the model only scales well if onboarding and support are highly standardised; otherwise service cost erodes gross margin.^{291 145}

- Most valuable segment: **The best customer cohort for durable margin is the upper mid-market and enterprise shipper base that is large enough to need integrated execution software, but not so large that every deployment becomes quasi-custom consulting.** This segment typically combines meaningful ACV, moderate implementation complexity, and strong appetite for add-on modules in customs, emissions, and planning. This is an inference based on Eurostat adoption gradients and the documented SME cost barriers.^{291 145}

8.3

Outsourcing and Platform Dependencies

Outsourcing is still expanding, but the form of value capture is changing from pure labor arbitrage toward digitally mediated control. Miebach's 2026 Logistics Outsourcing Study reports that around 60% of surveyed companies have already fully outsourced transportation services and 25% fully outsource warehouse operations, while the share of companies in Germany tendering warehouse services only every five years or every seven years or longer rose from 14% in 2023 to 22% in 2026, indicating longer relationship duration once outsourcing is embedded.²⁸⁴

- Acceleration pattern: The outsourcing trend is strongest in transport, where scale procurement, carrier management, and compliance complexity favour specialist providers. CSCMP's 2025 3PL study also reports continued demand for advanced technologies, digital capabilities, and visibility tools in shipper and 3PL relationships, reinforcing that outsourcing decisions increasingly include a technology stack decision.^{293 284}
- Dependency effect: As 3PLs and 4PL-style orchestrators take over execution, the software platform they standardise on becomes sticky for the underlying shipper. This raises platform loyalty for vendors embedded in outsourced operations because switching software may require retendering the service model itself, not just replacing an application.^{278 293}
- Limits to outsourcing: Outsourcing is not universally optimal. A 2025 Renault-related arXiv study on worldwide shipper transportation planning found that outsourcing was attractive in low-volume contexts, while large-scale networks benefited more from in-house planning, suggesting that the biggest shippers may still retain strategic planning authority even when execution is outsourced.²⁷³
- Vendor channel implication: Vendors selling through 3PLs can achieve efficient distribution and high seat density, but they risk concentration if a few large logistics providers control access to end customers. Vendors selling directly to shippers retain stronger account ownership, but may lose influence when the shipper later outsources operations to a provider with its own preferred stack.^{278 284}
- Investment conclusion: **The most attractive demand-side position is to be the default execution and data layer inside outsourced logistics relationships, especially in transport-heavy 3PL environments where retender cycles are lengthening and technology is part of the service contract.** By contrast, vendors that sit outside the outsourced operating model face a higher risk of disintermediation, even if their product is technically strong.^{284 293 273}

Investment Risk Register

9

Investment Risk Register

This section isolates the principal downside risks for investors in European logistics technology and links each risk to concrete underwriting and portfolio actions. The common pattern is that **margin durability deteriorates quickly when a vendor lacks workflow control, scale capital, or compliance infrastructure**, even if top-line growth initially appears attractive.^{316 24 158}

9.1

Commoditization and Funding Risks

Title — Investor risk register by exposure

RISK AREA	EVIDENCE SIGNAL	MARGIN IMPACT	MITIGATION PRIORITY
Spot brokerage commoditization	Funding to supply chain and logistics startups fell below \$6 billion across 741 announced deals in 2024, after the post-COVID boom unwound.	High take-rate pressure and cyclical EBITDA volatility.	Avoid principal-risk models without software attach.
Visibility point-solution commoditization	project44's 2026 positioning explicitly shifted from visibility toward AI agents and orchestration, implying tracking alone is no longer sufficient differentiation.	Stand-alone telemetry pricing compresses as features normalize.	Prefer system-of-action platforms with execution rights.
Series B and C scale-up gap	The EIB states that closing finance gaps for scale-ups is essential and that EU VC funds raise only 5% of global VC versus 52% in the US.	Growth stalls before sales scale and product breadth are achieved.	Underwrite follow-on capacity and syndicate quality.
Europe growth-capital scarcity in freight tech	Industry reporting in June 2026 notes that once freight-tech companies reach Series B or C, capital becomes harder to access because there are few European growth funds, while digital freight models also need working capital and local operating build-out.	Down-round risk, forced M&A, and weaker negotiating leverage.	Favor capital-efficient SaaS and buyers with dry powder.

Source: ^{310 249 316 312}

- Commoditization risk is highest in digital freight brokerage and basic visibility because both categories are increasingly judged on price, coverage, and workflow adjacency rather than on novel functionality. The 2025 funding retrenchment in supply-chain and logistics startups is consistent with investors re-pricing models whose revenue is still tied to freight cycles or easily replicated tracking layers.^{310 249}
- The strategic warning sign is when a vendor sells information without execution authority. project44's 2026 acquisition of LunaPath.ai and its broader move into agentic orchestration indicate that even category leaders view pure visibility as insufficient for durable pricing power, which is a negative read-through for subscale European point solutions.^{249 84}

- Funding risk is not just a macro backdrop issue but a direct determinant of competitive survival in Europe. The EIB's scale-up analysis says EU venture funds account for only 5% of global VC, and industry reporting in June 2026 specifically highlights a Series B and C bottleneck in freight tech, where companies often need both software investment and working capital to build carrier networks and local operations.^{316 312}
- For deal teams, the implication is that **the funding gap can create attractive entry points, but only where the target already has low-churn recurring revenue and does not require balance-sheet-heavy expansion to prove the model.** Capital scarcity is an opportunity in TMS, customs, and compliance software, but a hazard in brokerage, last-mile operations, and network businesses that still need subsidized growth.^{316 315 311}

9.2

Integration and Churn Risks

- ERP integration remains one of the most material execution risks in logistics software because the product rarely operates in isolation. OpenText states that company size, ERP platforms across geographies, and B2B or EDI development all determine scope and complexity, while its 2025 ERP guidance warns that poor synchronization between upstream order systems and downstream logistics or finance applications can cause transaction failures, inventory misallocation, and reporting breakdowns at go-live.^{158 326}
- In logistics tech, implementation failure is especially damaging because the software sits inside shipment execution, inventory movement, invoicing, and customer service. If integrations fail, the customer does not merely lose analytics visibility but risks operational disruption, which sharply increases churn probability and can also trigger delayed payments, scope disputes, and reputational damage. This is an inference supported by OpenText's description of go-live failure modes in ERP-linked logistics environments.^{326 158}
- The risk is amplified in Europe's mid-market because customers often run heterogeneous ERP estates across countries, business units, and acquired entities. OpenText's 2026 integration commentary argues that ERP, B2B integration, and AI must be treated as one connected transformation, which implies that vendors selling AI-led logistics workflows without robust integration architecture are likely understating delivery risk.^{328 158}
- Churn risk should therefore be underwritten less as a pure product issue and more as a services and implementation discipline issue. Vendors with prebuilt connectors, tested migration playbooks, externalized business rules, and strong partner ecosystems should sustain lower failed-deployment rates and better net retention than peers relying on bespoke integration work for each customer.^{326 158}
- The mitigation strategy for investors is clear. **Prefer targets where implementation has been industrialized into repeatable connectors, reference architectures, and measurable time-to-value milestones, and discount businesses whose growth depends on custom ERP remediation disguised as software revenue.**^{158 336}

9.3

Regulatory Compliance Burden

- Regulatory burden is rising faster than many smaller vendors' ability to absorb it. The Data Act has applied since 12 September 2025 and requires, among other things, open interfaces for

SaaS and software providers, machine-readable data export, and cloud-switching related obligations, while the Commission explicitly frames the regime as particularly relevant to SMEs facing unfair contractual terms and compliance frictions.^{24 330}

- The burden is not limited to one regulation. Recent arXiv work on EU AI Act operationalization shows that compliance now requires documentation, risk management, transparency, traceability, monitoring, and cross-functional ownership, while a separate 2026 survey finds many organizations still lack structured mechanisms to translate obligations into auditable requirements and lifecycle evidence.^{298 299}
- Smaller logistics tech vendors are disproportionately exposed because they must spread legal, security, product, and audit costs over a much smaller revenue base. The Commission's 2025 digital package states that simplification measures in data, AI, and cyber legislation are intended to reduce compliance costs and legal uncertainty, which is itself evidence that the baseline burden is material enough to warrant policy relief.^{335 24}
- There is also a direct margin effect. During the Data Act transition period through 12 January 2027, providers may still charge for certain switching and data-egress costs, but the long-term direction is toward lower lock-in and more interoperability, which can erode pricing power for vendors whose moat depends on closed data environments rather than superior workflow software.^{24 330}
- The investor response should be selective rather than defensive. **Back vendors that turn regulation into productized compliance modules, auditable data governance, and secure interoperability, and be cautious on subscale point solutions for which compliance spend is a fixed-cost drag rather than a monetizable feature set.**^{24 297 335}

10

CDD Verdict: Durable Margin Scorecard

10

CDD Verdict: Durable Margin Scorecard

This verdict section converts the prior analysis into an investor decision framework focused on margin durability, not category excitement. Across European logistics tech, the most durable margin pools remain concentrated in execution-critical software and network platforms with high recurring revenue, low ordinary-course churn, strong compliance attach, and growing AI-enabled workflow depth, while the weakest sit in transaction-led and hardware-first models where gross margin is structurally capped or cyclical.^{351 115 113 84 21}

The single most defensible margin thesis for a European logistics tech investor is to own platforms that sit inside daily transport, customs, warehouse, or carrier-execution workflows and can monetize the same data twice through compliance, audit, and AI modules. That combination produces the best mix of retention, pricing power, and operating leverage, and it is reinforced by the June 1, 2026 CountEmissionsEU standardization of transport emissions methodology, which increases the value of embedded reporting and optimization layers.^{21 115}

10.1

Subsector Margin Durability Analysis

Title — Margin durability ranking by subsector

SUBSECTOR	DURABILITY RANK	CORE EVIDENCE	INVESTOR READ-THROUGH
TMS, customs, and logistics execution SaaS	1	Descartes reported 46% adjusted EBITDA margin in FY2026 and indicated ordinary-course recurring revenue loss of only 5% to 7%; WiseTech reported 87% gross margin and 53% EBITDA margin in FY2025.	Best combination of recurring revenue, workflow lock-in, and compliance upsell.
WMS and warehouse execution software	2	Margin quality benefits from deep ERP and warehouse-process embedding; AI-led warehouse orchestration research shows meaningful efficiency gains when software controls task allocation and picking logic.	Strong durability when sold as system-of-action software rather than labor-light analytics.
Freight visibility and control tower platforms evolving into orchestration	3	project44 reported 34% year-over-year new ARR growth in Q1 FY27, 52% shipper new ARR growth, 67% of new ARR from multi-year deals, and about 300 bps NRR improvement.	Attractive if the platform owns carrier APIs, exception handling, and decision automation, not just tracking.
Warehouse automation and robotics platforms	4	Hybrid models can create recurring economics, but hardware, commissioning, and support dilute software-like margins; arXiv warehouse orchestration work supports efficiency upside but not pure SaaS economics.	Good businesses exist, but durability depends on software control and service annuities, not robot unit sales.

SUBSECTOR	DURABILITY RANK	CORE EVIDENCE	INVESTOR READ-THROUGH
Last-mile delivery tech	5	Margin durability varies sharply by revenue model; software layers can be sticky, but operational exposure, parcel pricing pressure, and local competition limit consistency.	Selective only in dense verticals such as healthcare, returns, or regulated delivery workflows.
Digital freight brokerage platforms	6	Transaction-led economics remain exposed to freight cycles and take-rate compression; recent arXiv work on dynamic pricing highlights structural instability in fragmented LTL markets.	Lowest durability and highest risk of mistaking volume growth for durable margin.

Source: ^{351 115 113 84 1 101}

- TMS, customs, and execution SaaS rank first because they combine the cleanest public margin evidence with the strongest switching-cost architecture. Descartes' FY2026 adjusted EBITDA margin of 46% and its disclosed 5% to 7% ordinary-course recurring revenue loss provide a rare current benchmark for what durable logistics software economics look like in practice.^{351 115}
- WMS and warehouse execution software rank second because they are deeply embedded in inventory, labor, and fulfillment workflows, but their margin quality is more implementation-sensitive than transport execution software. The category becomes especially attractive when AI and orchestration are layered into the operating core rather than sold as adjacent analytics.^{1 148}
- Visibility and control towers rank third, but only where they have crossed from telemetry into action. project44's May 18, 2026 disclosure of stronger multi-year deal mix and improved NRR is the clearest current signal that orchestration can convert a historically vulnerable category into a more durable recurring-revenue model.⁸⁴
- Warehouse automation and robotics rank below software-first categories because hardware and deployment intensity cap margin scalability even when customer stickiness is high. The best assets in this segment are software-led automation platforms with maintenance and optimization annuities, not equipment vendors dependent on project revenue.^{1 148}
- Last-mile tech sits in the middle-to-lower tier because local density, labor cost, and parcel pricing create persistent margin pressure. The category improves materially only when the product is tied to regulated delivery, returns, or enterprise parcel orchestration rather than generic dispatch and routing.
- Digital freight brokerage ranks last because the model remains structurally exposed to freight-rate cycles, carrier liquidity, and take-rate competition. Even where pricing engines improve matching efficiency, the underlying economics are still closer to market intermediation than to high-retention software annuities.^{101 144}

10.2

Investment Quality Classification

- TMS, customs, and execution SaaS: **Investment-grade.** The evidence base is strongest here because current public leaders show software-like gross and EBITDA margins, low recurring

revenue attrition, and clear regulatory upsell paths into customs, trade compliance, and emissions reporting.^{115 113 21}

- WMS and warehouse execution software: **Investment-grade**. The category benefits from high workflow embeddedness and meaningful switching friction, and AI-led orchestration can improve labor productivity and throughput. The main underwriting test is whether the target sells repeatable software and services or hides custom implementation work inside reported software growth.^{1 148}
- Freight visibility and control tower platforms: **Conditional investment-grade**. This is investment-grade only when the platform has carrier network density, multi-year contracts, improving NRR, and a credible move into orchestration or API infrastructure. Pure milestone-tracking vendors remain vulnerable to feature commoditization and should be treated as value-trap candidates.⁸⁴
- Warehouse automation and robotics: **Selective investment-grade, otherwise value-trap risk**. The category can produce durable revenue where software, maintenance, and installed-base optimization dominate lifetime value, but hardware-first models often overstate defensibility because customer stickiness does not automatically translate into software-like margins. This is an inference from the technical evidence on orchestration value and the broader margin gap between software and deployment-heavy models established earlier in the report.^{1 148}
- Last-mile delivery tech: **Mostly value-trap, with niche exceptions**. Generic last-mile software faces local competition, limited pricing power, and exposure to parcel and labor economics. The exceptions are verticalized products in healthcare, returns, or compliance-heavy delivery environments where service complexity creates real switching costs.
- Digital freight brokerage: **Value-trap**. The category can show attractive top-line growth in favorable freight conditions, but margin durability is weak because revenue quality is transactional, take rates are contested, and the model remains exposed to macro and network liquidity shocks.^{101 144}
- Overall classification rule: **Investment-grade subsectors are those where recurring revenue, workflow lock-in, and compliance or AI attach reinforce each other. Value-traps are those where growth depends on freight volume, hardware deployment, or point-solution novelty without a durable system-of-action position.**^{115 84 21}

10.3

Strategic Recommendations

- Buy: Prioritize TMS, customs, trade-compliance, and transport execution platforms with proven recurring revenue, low churn, and attach potential for emissions, audit, and AI modules. These assets offer the clearest path to software-style margin durability and remain the best fit for platform-plus-bolt-on strategies.^{115 113 21}
- Buy: Selectively acquire WMS and warehouse execution software where the product already controls labor, slotting, picking, or automation orchestration. The investment case is strongest when the target can expand into warehouse AI, returns, or robotics control without taking principal hardware risk.^{1 346}
- Build: Develop compliance-by-design modules in carbon accounting, customs documentation, freight audit, and secure data-sharing on top of existing execution platforms. The June 1, 2026 CountEmissionsEU framework creates a timely product wedge because customers now need standardized, auditable transport emissions outputs across modes.²¹

- Partner: In visibility and control towers, partner unless the asset already owns carrier connectivity and execution rights. Network density matters, but the category is increasingly winner-take-most, so subscale entrants should use partnerships with 3PLs, TMS vendors, or carrier networks rather than attempt greenfield network creation.^{84 146}
- Partner or buy tuck-ins: In warehouse automation, favor orchestration software, maintenance layers, and installed-base analytics over robot OEM exposure. Where physical automation capability is required, partnership with integrators or OEMs is usually superior to owning low-margin hardware manufacturing.
- Avoid building from scratch: Do not build generic digital freight brokerage or undifferentiated last-mile platforms unless there is a captive demand channel, a regulated niche, or a proprietary data advantage. These areas are the most likely to absorb capital without producing durable margin expansion.¹⁰¹
- Final strategic implication: **The best place to build equity value in European logistics tech is the software control layer that turns execution data into mandatory workflows, then monetizes adjacent compliance and AI services at high incremental margin.** That is the narrowest and most defensible route to durable margin in the sector as of August 25, 2026.^{115 84 21}

Sources & References

11

Sources & References

This section consolidates the principal source base used across the report and distinguishes between primary evidence, contextual references, and citation mechanics. The emphasis is on investor-grade traceability, with priority given to company filings, EU regulatory texts, Eurostat datasets, and named market or transaction reports that were current as of August 25, 2026.

11.1

Primary Data Sources

- Company financial disclosures: Core operating, margin, strategy, and M&A evidence was drawn from current annual reports, investor presentations, and regulatory filings for DSV, DHL Group, Kuehne+Nagel, Maersk, Descartes, WiseTech Global, and Ocado. These sources were treated as primary evidence for revenue mix, profitability, strategic positioning, and disclosed acquisition activity because they are issuer-authenticated and current to 2025 or 2026.^{69 71 94 96 115 113 108}
- EU regulatory and policy documents: Regulatory analysis relied primarily on European Commission and EUR-Lex materials covering CountEmissionsEU, the EU ETS maritime regime, the Data Act, CBAM, and eFTI. These documents were the authoritative basis for effective dates, scope, and compliance implications because they define the legal and policy framework directly.^{21 160 24 202 203 204 205}
- Official statistical sources: Market context, enterprise digitisation, AI adoption, and e-business penetration were anchored in Eurostat and related EU statistical publications. These sources were used where the report required comparable cross-country evidence rather than vendor marketing estimates.^{22 361 362 366}
- Market and transaction datasets: Third-party market sizing and valuation references were used selectively where primary public disclosures did not provide Europe-specific category estimates or private-market comp ranges. These sources were treated as directional benchmarks rather than as sole evidence where methodology transparency was limited.^{363 258 264 255}
- Technical and academic evidence: arXiv papers were used to support technology-readiness, digital twin, warehouse orchestration, and network optimization arguments where public company disclosures alone were insufficient. They were not used as substitutes for market sizing or legal interpretation, but as technical corroboration for feasibility and implementation direction.^{148 173 101 144}

11.2

Secondary Data References

- Industry context and adoption studies: Supporting context on AI adoption, outsourcing, and procurement behavior was supplemented with consulting, trade-association, and specialist industry research where official statistics or issuer filings did not address workflow-level buying behavior. These references were used to frame demand-side interpretation rather than to override primary evidence.^{178 189 293 284}
- Company press releases and investor communications: For recent M&A, product launches, and private-company operating signals, the report used company press releases where no audited

filing equivalent was available. This was especially relevant for project44, DHL transaction commentary, and Descartes acquisition announcements.^{84 249 252 250}

- Market sizing and category estimates: Where Europe-only subsector sizing was not available from official statistics, the report referenced named market research providers for warehouse automation, robotics, and WMS estimates. These figures were used with explicit caution because methodology and segmentation can vary materially across providers.^{46 44 32}
- Broader macro and policy context: Additional European Commission and Eurostat materials were used to support macro, trade, and strategic-foresight framing around resilience, nearshoring, and geopolitical risk. These references provided contextual support for the investment case but were not the sole basis for subsector conclusions.^{31 229 369}
- Integration and implementation references: OpenText materials were used as supporting evidence on ERP, B2B integration, and go-live complexity because they describe practical implementation failure modes relevant to logistics software churn and deployment risk. These were treated as practitioner references rather than neutral market statistics.^{158 163 326}

11.3

Citation Details

- Source hierarchy: The report followed a strict evidence hierarchy. First priority was given to issuer filings, SEC exhibits, annual reports, EU regulations, European Commission notices, and Eurostat datasets; second priority to named transaction reports and consulting studies; third priority to company press releases and technical papers for product, workflow, or feasibility evidence.^{115 21 22}
- Use by evidence type: Financial metrics such as gross margin, EBITDA margin, recurring revenue mix, and disclosed churn proxies were cited from company filings or investor materials. Regulatory dates and obligations were cited from European Commission or EUR-Lex pages, while valuation ranges and private-market read-throughs were cited from named M&A reports where public filings did not disclose transaction multiples.^{113 160 258}
- Treatment of private-company claims: Where private companies such as project44 disclosed ARR growth, NRR movement, carrier-network scale, or product launches, those claims were cited directly to the company announcement and interpreted as management-reported rather than independently audited figures. This distinction is important for diligence because such data is useful for directional analysis but should be re-verified in deal workstreams.^{84 88}
- Treatment of third-party market estimates: Market research figures for WMS, warehouse automation, and robotics were cited only where no better public Europe-specific series was available, and the report explicitly avoided presenting them as precision-grade facts when methodologies diverged. Readers should therefore treat these estimates as triangulation inputs rather than audited market totals.^{46 44 32}
- Validation note: All references listed above were selected because they were accessible, current, and directly relevant to the report's claims as of August 25, 2026. Where the public record did not support a precise Europe-only metric, the report stated that limitation explicitly rather than inferring unsupported precision.^{264 361 148}



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